UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE POLITICHE, ECONOMICHE E SOCIALI

Convolutional Neural Network Architectures for Rock–Paper–Scissors Image Classification

End-Of-Course Project

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Data Science for Economics (Master's Degree)
II Year

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February 28, 2026

Abstract

This project investigates the impact of architectural complexity on the performance and robustness of convolutional neural networks for image classification, using the Rock–Paper–Scissors hand gesture task as a controlled case study. Three CNN architectures with progressively increasing capacity are designed and compared under identical training conditions, including consistent data preprocessing, augmentation, and optimization strategies. Model selection is guided by a constrained hyperparameter tuning procedure based on validation accuracy. Beyond standard in-distribution evaluation, particular emphasis is placed on generalization under domain shift through testing on an external dataset of unconstrained, real-world images. The experimental results reveal a clear relationship between model capacity and in-distribution performance, while also highlighting that strong validation accuracy does not necessarily guarantee robustness to distributional changes. The findings underscore the importance of evaluating neural network models beyond held-out validation data when assessing their reliability in real-world deployment scenarios.

1 Introduction

Convolutional Neural Networks (CNNs) have become a standard approach for image classification tasks due to their ability to automatically learn hierarchical feature representations from raw visual data. In recent years, CNN-based models have demonstrated strong performance across a wide range of computer vision applications, from object recognition to gesture and action classification.

Despite these successes, the relationship between architectural complexity and model generalization remains a central issue in deep learning. While increasing model capacity often leads to improved performance on held-out validation data, it does not necessarily translate into robust behaviour when models are exposed to distributional shifts or unconstrained real-world inputs. This challenge is particularly relevant for vision tasks involving variability in lighting, backgrounds, and acquisition conditions, such as hand gesture recognition.

Within this context, the present study uses the Rock–Paper–Scissors image classification task as a controlled benchmark to analyse how different CNN architectures behave as model complexity increases. By comparing multiple architectures under a consistent experimental protocol, the project aims to provide insight into how architectural choices affect learning dynamics, error structure, and generalization properties under both standard and shifted data distributions.

The remainder of the report is organized as follows. Section 2 describes the dataset and preprocessing pipeline, Section 3 presents the model architectures, Section 4 reports the experimental results, and Section 5 discusses generalization under domain shift.

2 Data Exploration and Preprocessing

2.1 Dataset Description and Exploratory Analysis

The dataset used in this project was obtained from a publicly available Kaggle repository (`https://www.kaggle.com/datasets/drgfreeman/rockpaperscissors`) containing images of hand gestures for the Rock–Paper–Scissors game. The dataset consists of a total of 2,188 RGB images, evenly distributed across the three target classes: rock (726 images), paper (710 images), and scissors (752 images). All images were captured against a green background under relatively consistent lighting and white balance conditions, resulting in a visually homogeneous and well-curated dataset.

Each image is stored in PNG format with an original resolution of 300 × 200 pixels and is organized into class-specific subdirectories, enabling a straightforward mapping between file structure and class labels. An initial exploratory analysis confirmed the near-balanced class distribution and the absence of missing or corrupted files. Random visual inspection of samples from each class further revealed that the hand gestures are generally well-centered and clearly distinguishable, with limited intra-class variability compared to real-world acquisition scenarios. This level of consistency makes the dataset particularly suitable for controlled experiments focused on architectural comparisons rather than robustness to extreme visual noise.

To avoid any form of data leakage, external datasets are introduced only at a later stage of the analysis. These datasets are used exclusively for evaluation and play no role in model training or model selection, allowing generalization to be assessed independently of the training distribution.

2.2 Train – Validation Split

The dataset was split into training and validation subsets using a stratified sampling strategy, with 80% of the images allocated to training and 20% reserved for validation. Stratification was performed at the file level to preserve the original class proportions in both splits and to guarantee that all classes are represented in each subset.

A fixed random seed was used to ensure reproducibility of the split across runs. The validation set was held out entirely from training and used exclusively for model selection, hyperparameter tuning, and performance evaluation. This splitting strategy provides a reliable estimate of in-distribution generalization while maintaining methodological transparency and repeatability.

2.3 Image Preprocessing: Resizing and Normalization

After splitting the dataset into training and validation subsets, a common preprocessing pipeline was applied to all images: to ensure compatibility with convolutional neural network architectures and to standardize the input space, images were resized to a fixed square resolution of 96 × 96 pixels. Resizing was performed using bilinear interpolation with antialiasing, preserving the essential spatial structure of the hand gestures while reducing computational cost and memory requirements.

Following resizing, pixel intensities were converted to floating-point format and normalized to the [0, 1] range, with the aim of facilitating stable and efficient optimization by preventing large gradient magnitudes during training. This normalization step was applied directly within the data pipeline rather than as a dedicated rescaling layer inside the models, ensuring consistent preprocessing across all architectures and simplifying model definitions.

2.4 Data Augmentation

Although the dataset is visually consistent and relatively clean, light data augmentation was applied during training to encourage better generalization and to reduce sensitivity to small spatial variations. Augmentation operations were deliberately kept mild to avoid introducing unrealistic distortions, given the already constrained nature of the dataset.

Specifically, the training pipeline includes random horizontal flips, small rotations, zooming, translations, and slight contrast adjustments, applied exclusively to the training set to ensure that validation data remains untouched and representative of the original distribution. These transformations are applied on-the-fly at batch loading time, meaning that each epoch exposes the model to a different randomized version of the same training samples. This strategy allows the models to learn more robust feature representations while preserving a fair and unbiased evaluation protocol.

3 CNN Architecture and Training

3.1 Design of CNN Architectures with Incremental Complexity

The three convolutional neural network architectures evaluated in this study were designed to form a controlled complexity hierarchy, in which representational capacity, depth, and regularization mechanisms are progressively increased while keeping the input resolution and training protocol fixed.

Formally, each model defines a mapping $f(\mathbf{x}_0; \Theta)$ from an input image $\mathbf{x}_0 \in \mathbb{R}^{H\times W\times 3}$ to class probabilities via a softmax layer.

To make the concept of incremental increase in model capacity more explicit, Table 1 reports the number of trainable parameters and effective model size for each architecture.

Table 1: *Number of trainable parameters and model size across architectures.*

Model	Trainable parameters	Model size (trainable)
Model A	2,814	11 KB
Model B	249,331	974 KB
Model C	45,523	178 KB

Model A. Model A is a deliberately lightweight baseline architecture based on depthwise-separable convolutions. It consists of three separable convolutional layers with increasing channel depth, where the first two are followed by max-pooling for spatial downsampling:

$$\mathbf{x}_{l+1} = \sigma(\texttt{SepConv}(\mathbf{x}_l)), \qquad \mathbf{x}_{l+2} = \texttt{MaxPool}(\mathbf{x}_{l+1})$$

After the final convolution, global average pooling (GAP) is applied:

$$\mathbf{z} = \texttt{GAP}(\mathbf{x}_L),$$

followed by dropout and a small dense classifier:

$$\hat{\mathbf{y}} = \texttt{Softmax}\big(W_2\,\texttt{ReLU}(W_1\,\texttt{Dropout}(\mathbf{z}))\big).$$

The use of separable convolutions significantly reduces model capacity (2,814 trainable parameters; see Table 1), enforcing a strong inductive bias toward coarse spatial structure. Combined with global average pooling, this limits memorization of localized patterns, so the model primarily captures global shape cues rather than fine-grained details.

Model B. Model B is a conventional shallow CNN with two convolution–pooling blocks followed by a fully connected classifier. Using standard (valid) convolutions:

$$\mathbf{x}_{l+1} = \sigma(\texttt{Conv2D}(\mathbf{x}_l)), \qquad \mathbf{x}_{l+2} = \texttt{MaxPool}(\mathbf{x}_{l+1}).$$

After the second block, feature maps are flattened,

$$\mathbf{z} = \texttt{Flatten}(\mathbf{x}_L),$$

and fed to a small dense head:

$$\hat{\mathbf{y}} = \texttt{Softmax}(W_2\,\texttt{ReLU}(W_1\mathbf{z})).$$

Compared to model A, this design increases capacity (249,331 parameters) by preserving spatial degrees of freedom via flattening, at the cost of weaker implicit spatial regularization.

Model C. Model C is a deeper architecture designed for stable training under limited data. With 45,523 trainable parameters, it targets a favorable capacity–efficiency trade-off. The network begins with a strided convolutional stem,

$$\mathbf{x}_1 = \texttt{Conv}_{s=2}(\mathbf{x}_0),$$

followed by staged depthwise-separable blocks with layer normalization and ReLU. Let $F(\mathbf{x};\theta)$ denote one such block (depthwise 3×3 with stride s, pointwise 1×1, LN, ReLU). Residual connections are applied only when shapes match:

$$\mathbf{y} = \begin{cases} \mathbf{x} + F(\mathbf{x};\theta), & s = 1 \ \wedge \ C_{\text{in}} = C_{\text{out}}, \\ F(\mathbf{x};\theta), & \text{otherwise.} \end{cases}$$

No projection shortcut is used when dimensions change, keeping blocks lightweight. Spatial resolution is reduced via stride-2 depthwise blocks, yielding feature maps at 48×48, 24×24, and 12×12, followed by a 1×1 bottleneck convolution. Global average pooling produces

$$\mathbf{z} = \texttt{GAP}(\mathbf{x}_L),$$

and the classifier head is

$$\hat{\mathbf{y}} = \texttt{Softmax}\big(W_2\,\texttt{Dropout}(\texttt{ReLU}(W_1\mathbf{z}))\big),$$

with an optional bias initialization using log-prior class probabilities.

3.2 Architectural Design Choices and Technical Rationale

While the previous subsection described the structural composition of each architecture, we now discuss the underlying design rationale and regularization choices that ensure training stability and fair comparison.

First, depthwise separable convolutions, already introduced in Models A and C, are primarily motivated by robustness under limited data conditions. In Model C, separable convolutions are implemented explicitly via depthwise and pointwise operations and are combined with conditional residual connections. These residual shortcuts are applied only when spatial resolution and channel dimensionality match, improving gradient flow without introducing projection parameters.

Second, Layer Normalization (LN) is employed instead of Batch Normalization in Model C. Since training is performed with relatively small batch sizes and with data augmentation enabled, LN provides more stable normalization statistics that are independent of batch composition. This choice improves convergence consistency and reduces sensitivity to batch-size variations.

Beyond their structural description, the downsampling mechanisms differ systematically across architectures. Models A and B rely on MaxPooling layers for progressive spatial reduction, whereas Model C performs downsampling through strided convolutions (both in the initial stem and in depthwise blocks). This design allows Model C to learn feature extraction and resolution reduction jointly, preserving richer intermediate representations.

Consistent with their parameter-efficient design, Models A and C employ Global Average Pooling at the classification stage. In contrast, Model B uses flattening, resulting in a higher-capacity fully connected head.

Label smoothing is applied uniformly across all models through the categorical cross-entropy loss, preventing overconfident predictions and improving robustness under domain shift. In Model C, the optional initialization of output biases using empirical log-prior class probabilities further stabilizes early training dynamics, particularly in the presence of mild class imbalance.

Across all architectures, optimization is performed using the Adam optimizer with default β_1 and β_2 coefficients. Model A is trained with a learning rate of 10^{-3}, Model B uses the default Adam learning rate, and Model C adopts a reduced learning rate of 3×10^{-4} to improve optimization stability in the deeper architecture.

Training is regularized through early stopping monitored on validation accuracy with restoration of the best-performing weights and an adaptive learning-rate schedule (ReduceLROnPlateau), which reduces the learning rate when validation performance fails to improve for a predefined number of epochs. Together, these mechanisms promote stable convergence and mitigate overfitting while limiting unnecessary computation once validation performance saturates.

3.3 Hyperparameter Tuning and Final Model Selection

Hyperparameter tuning is carried out through a constrained grid search applied consistently across all three architectures (Models A, B, and C), exploring a small set of high-impact training hyperparameters: learning rate, batch size, and the use of data augmentation. For each architecture, all combinations in the grid are trained using the same stratified train–validation split, and model selection is based on validation accuracy computed on the held-out validation set. Early stopping is employed during each run to prevent overfitting and to limit unnecessary computation once validation performance plateaued, enabling an efficient yet systematic exploration of the search space. All trials are logged in a dedicated results table (`tuning_results.csv`), ensuring full reproducibility and allowing direct comparison across architectures under matched tuning conditions. The best configurations identified for each model are summarized in Table 2.

Table 2: *Best tuning configurations per model, reporting the best run*

Model	Learning rate	Batch size	Augment	Val. accuracy
A	1×10^{-3}	32	Yes	0.6644
B	3×10^{-4}	32	No	0.9658
B	1×10^{-3}	32	Yes	0.9658
C	3×10^{-4}	16	No	0.9954

The best configuration identified for each architecture is subsequently used for a clean final training run with checkpointing and evaluation artefacts generated accordingly.

Section 3 presents a detailed comparison of validation performance across architectures and the resulting model ranking. The overall best-performing model is then selected for external generalization tests on the two additional datasets.

4 Evaluation and Analysis

Building on the architectural comparison in Section 2, this section examines how increasing model complexity translates into empirical performance differences. We report a selected subset of evaluation outputs that are most informative for assessing predictive performance, training stability, and generalization. In particular, we compare architectures using validation metrics (accuracy, precision, recall, F1-score), analyze training dynamics through loss and accuracy curves, inspect misclassified examples to identify systematic error patterns, and integrate these elements to diagnose overfitting or underfitting. Additional intermediate diagnostics used during development are omitted for brevity, as they do not provide further interpretative value.

4.1 Performance Metrics

Model performance was assessed using standard multi-class classification metrics, including accuracy, precision, recall, and F1-score, computed on a held-out validation set. Each metric captures a different and complementary aspect of predictive behaviour, allowing a comprehensive view of both predictive quality and learning behaviour and a nuanced comparison across architectures beyond accuracy alone. While accuracy summarizes overall correctness, precision and recall capture complementary aspects of class-specific performance: precision reflects the model's ability to avoid false positive assignments, whereas recall measures its capacity to correctly retrieve all instances of a given class. The F1-score balances these two dimensions and is particularly informative when comparing models with different error profiles. Support values confirm that all metrics are computed on a nearly balanced validation set, ensuring that macro-averaged scores are meaningful and not driven by class imbalance.

Table 3: *Validation performance of the best tuned configuration for each architecture. Metrics are macro-averaged over classes (3 decimal precision).*

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Model A (best)	0.651	0.651	0.650	0.645
Model B (best)	0.961	0.962	0.961	0.961
Model C (best)	0.993	0.993	0.993	0.993

As shown in Table 3, a clear and substantial improvement emerges across architectures, with a pronounced performance gap between the baseline model and the two more expressive networks.

The baseline architecture (Model A) achieves a validation accuracy of 0.651 and a macro F1-score of 0.645, indicating that it fails to reliably separate the three gesture classes even under in-distribution conditions. The close alignment between macro precision (0.651) and recall (0.650) suggests that performance limitations are not driven by a specific bias toward false positives or false negatives, but rather by an overall inability to extract sufficiently discriminative representations. In practical terms, this behaviour corresponds to frequent class confusion and unstable decision boundaries, consistent with limited representational capacity.

Model B marks a decisive qualitative shift in performance. Validation accuracy rises sharply to 0.961, with macro precision, recall, and F1-score all exceeding 0.96. This substantial gain relative to Model A indicates that the architectural modifications introduced in Model B enable the learning of far more expressive and class-separable features. The near-perfect balance between precision and recall across classes suggests that errors are both rare and symmetrically distributed, pointing to stable and well-calibrated decision regions.

The most expressive architecture (Model C) further consolidates these gains, achieving a validation accuracy of 0.993 and macro F1-score of 0.993. The marginal improvement over Model B reflects a transition from strong to near-saturated performance within the validation distribution. At this level, residual misclassifications become sparse and largely non-systematic. Importantly, the concurrent improvement in all reported metrics indicates that performance gains are not the result of a precision–recall trade-off, but rather of a genuine and uniform reduction in classification errors.

Taken together, these results reveal a two-stage progression in model capability. The transition from Model A to Model B produces a dramatic improvement driven by increased representational power, while the transition from Model B to Model C yields a more incremental refinement toward near-complete class separability. This progression provides a principled and empirically grounded justification for selecting Model C as the final architecture for subsequent generalization and robustness analyses.

Prediction confidence on the validation set. In addition to accuracy-based metrics, the distribution of predicted probabilities provides a coarse proxy for predictive confidence. Although this does not constitute a formal calibration assessment, it offers a simple summary of how confident each model is in its predictions. On the validation set, the mean maximum softmax probability increases markedly with model expressiveness (Model A: 0.44, Model B: 0.90, Model C: 0.95). This trend mirrors the improvement observed in classification accuracy and indicates that higher-performing architectures not only make fewer errors but also assign substantially higher confidence to their decisions.

The relatively low average confidence of Model A is consistent with its limited discriminative capacity and frequent class confusion, whereas Models B and C operate in a high-confidence regime with reduced uncertainty. Importantly, class-averaged predicted probabilities remain approximately balanced across classes, suggesting that improvements are not driven by degenerate class priors but by genuinely more separable representations. For completeness, validation-set confidence diagnostics (mean and standard deviation of the maximum softmax probability) are reported in Appendix A.

4.2 Training Dynamics

Training dynamics were analysed by jointly examining the evolution of loss and accuracy on both the training and validation sets. Figure 1 provides insight into how each architecture learns over time, whether performance improvements arise from genuine feature learning or from overfitting, and how effectively optimization translates into generalization within the validation distribution.

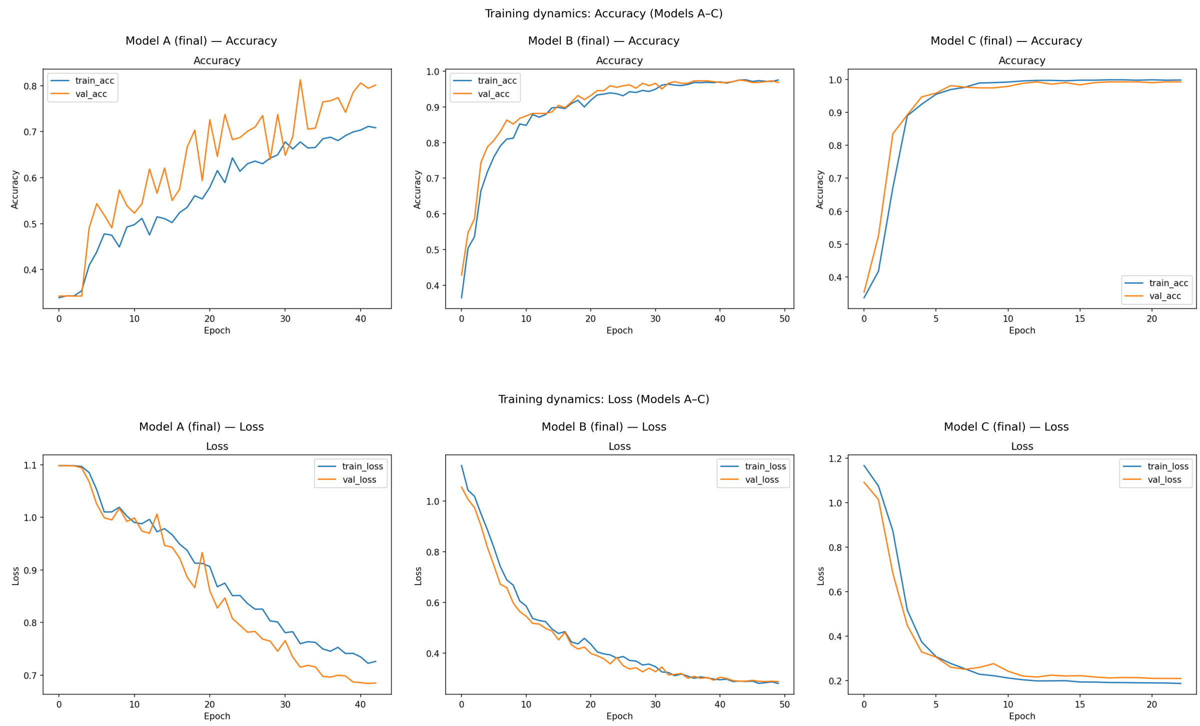

Figure 1: *Training and validation accuracy (top) and loss (bottom) curves for the three final architectures.*

For the baseline architecture A, both training and validation loss decrease steadily and approach a plateau; accuracy improves slowly and remains substantially below models B/C. Validation accuracy is noticeably noisier and often slightly higher than training accuracy, consistent with a limited-capacity model trained under augmentation/ regularization rather than with memorization-driven overfitting. The absence of a widening gap in the overfitting direction (i.e., training performance substantially exceeding validation performance) indicates that the model does not overfit the data. Instead, the early stagnation of performance suggests limited representational capacity: the model is unable to further reduce classification error even on the training set. In practical terms, this behaviour is characteristic of mild underfitting, where the architecture captures only coarse visual patterns and fails to exploit additional training iterations to improve class discrimination.

Model B exhibits a markedly different learning trajectory. Loss decreases steadily across epochs for both training and validation sets, accompanied by a rapid and sustained increase in accuracy. The close alignment between training and validation curves throughout the optimization process indicates stable learning and efficient use of model capacity. From a practical perspective, this pattern suggests that the architectural enhancements introduced in model B enable the network to learn more expressive features without sacrificing generalization, leading to consistent improvements in both error reduction and predictive accuracy.

Model C shows similarly smooth and stable convergence, but with more rapid early error reduction and lower final loss values. Training and validation curves remain tightly coupled, with only a minimal and non-systematic generalization gap (the training loss appears slightly below validation loss at convergence). This pattern indicates that the increased model capacity does not translate into late-stage divergence between training and validation performance. Instead, the adopted training configuration appears sufficient to maintain stable generalization within the validation distribution. The model is therefore able to leverage its higher representational capacity without exhibiting clear signs of memorization-driven overfitting. In practical terms, this behaviour corresponds to rapid acquisition of highly discriminative features and convergence toward a stable low-loss regime rather than unstable or capacity-induced degradation.

Overall, the progression of training dynamics across models mirrors the performance trends observed in the classification metrics. Model A is constrained by insufficient capacity, model B achieves an effective balance between learning and generalization, and model C combines high expressiveness with stable optimization. These dynamics strongly suggest that the observed performance differences are rooted in structural properties of the architectures rather than in optimization artefacts or training instability.

4.3 Analysis of Misclassified Examples

When informative, a qualitative analysis of misclassified examples was conducted to identify systematic error patterns and to complement the quantitative evaluation provided by aggregate performance metrics. In this context, *misclassified samples* are defined as validation images for which the predicted class label differs from the ground-truth label. For each such instance, the predicted class confidence is computed as the maximum softmax probability associated with the model's output. This allows misclassifications to be inspected not only in terms of correctness, but also in terms of model confidence.

To provide a compact overview of class-level error patterns before moving to instance-level inspection, Figure 2 reports the validation confusion matrices for the three final architectures. These matrices summarize how misclassifications are distributed across classes and offer a global complement to the analysis presented below.

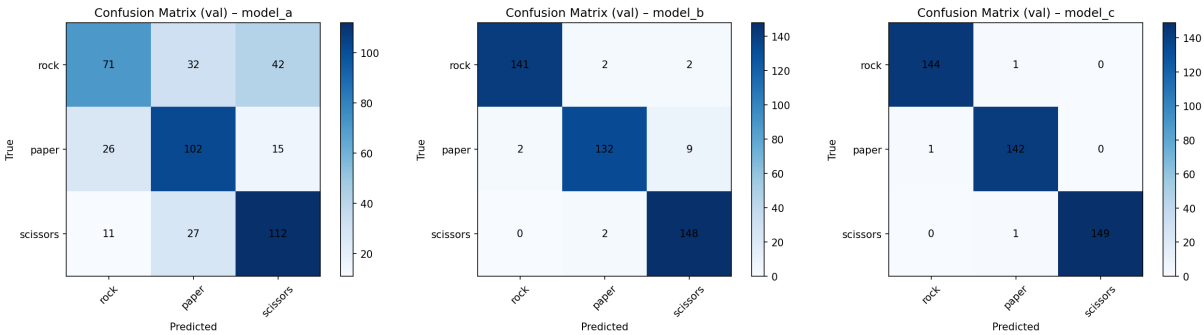

Figure 2: *Validation confusion matrices for the three final architectures (models A–C).*

Misclassified samples are ranked according to their prediction confidence and a fixed number of the most confident errors is selected for qualitative inspection. This design choice focuses the analysis on the most informative failure cases, namely those in which the model assigns high confidence to an incorrect prediction. Such errors are particularly relevant, as they reveal systematic confusions in the learned representation rather than borderline or noisy instances. Representative misclassified examples for each final architecture are shown in Apendix B, Figures 3, 4 and 5, ordered by decreasing prediction confidence.

The qualitative patterns observed differ substantially across architectures.

For Model A, misclassifications are frequent and distributed across multiple class pairs rather than concentrated in a single dominant confusion. The confusion matrix shows substantial leakage from rock into both paper (32 instances) and scissors (42 instances), as well as notable confusion between scissors and paper (27 instances). Errors also occur in the paper-to-rock direction (26 instances), indicating generally unstable decision boundaries across all three classes.

The grid of most confident misclassified samples further illustrates this behaviour. Among the highest-confidence errors, several instances—particularly true rock and true scissors samples—are predicted as paper with moderate confidence values in the range 0.6–0.8. While paper is not the sole source of confusion in the overall distribution, it appears prominently among the most confident mistakes, suggesting asymmetric confidence calibration across classes.

Overall, this pattern supports the underfitting diagnosis: Model A relies primarily on coarse global shape cues and fails to consistently capture the fine-grained structural differences between closed fists and extended finger configurations. The resulting error structure reflects limited representational capacity and insufficient class separation.

For Model B, misclassifications are substantially fewer and more localized than in Model A. The confusion matrix indicates only a small number of errors per class, with most off-diagonal entries remaining close to zero. The only slightly more visible pattern is a limited confusion of paper instances predicted as scissors (9 cases), while confusions involving rock are rare and largely symmetric.

The grid of most confident errors confirms that mistakes typically arise in visually ambiguous cases, such as partially extended fingers or transitional poses. Although a few errors occur with relatively high confidence (notably some paper–scissors confusions), these cases remain isolated and do not form a systematic failure mode.

Compared to Model A, the learned representation appears significantly more stable and class-separable, as reflected in the sharp reduction of off-diagonal counts and the substantial improvement in overall validation accuracy.

For Model C, misclassifications are extremely rare and highly localized. The confusion matrix is nearly diagonal, with only isolated off-diagonal entries and no dominant confusion pattern.

The few errors visible in the grid correspond to borderline or visually ambiguous cases, such as partially extended fingers or subtle pose variations. Although some of these mistakes occur with relatively high confidence (e.g., paper misclassified as rock or scissors predicted as paper), they remain isolated events rather than manifestations of a systematic structural bias toward a specific incorrect class.

Overall, the qualitative evidence aligns with the near-saturated validation metrics: residual errors appear to stem primarily from intrinsic visual ambiguity in individual samples rather than from insufficient representational capacity. This contrasts sharply with Model A and confirms that the increased architectural depth and normalization strategy effectively enhance class separability.

4.4 Overfitting and Underfitting Assessment

The presence or absence of overfitting and underfitting was assessed by jointly analysing quantitative performance metrics, training and validation learning curves, and the qualitative structure of classification errors. This combined perspective allows potential optimization artefacts to be distinguished from genuine limitations in model capacity.

For the baseline architecture (Model A), clear evidence of underfitting emerges. Training accuracy saturates at relatively low levels and does not substantially improve with additional epochs, while training and validation loss curves remain closely aligned throughout optimization. This behaviour indicates that the model fails to reduce error even on the training data, effectively ruling out memorization-driven overfitting and instead pointing to insufficient representational capacity.

The qualitative inspection of misclassified examples further supports this diagnosis. Errors are frequent and distributed across multiple class pairs, with substantial confusion between rock, paper, and scissors rather than a single dominant failure mode. The absence of sharply defined class boundaries, together with moderate confidence levels for many incorrect predictions, suggests that the model relies primarily on coarse global cues and fails to capture the fine-grained structural differences between gesture configurations. This pattern is consistent with limited representational capacity rather than noise sensitivity.

In contrast, no evidence of overfitting is observed for the more expressive architectures. For both models B and C, training and validation accuracy curves evolve in parallel, and validation loss does not exhibit late-stage divergence. The generalization gap remains small and stable, indicating that performance improvements are not driven by memorization of the training set but by the learning of increasingly discriminative representations.

Model B operates in a regime of effective capacity: it substantially reduces classification error while maintaining stable convergence and a balanced error profile across classes. Model C further increases representational power without inducing instability or widening the generalization gap. Although its performance approaches saturation on the validation set, the absence of divergence between training and validation metrics suggests that the model remains well-regularized under in-distribution conditions rather than over-parameterized.

Taken together, these findings indicate a progressive transition from capacity-limited underfitting (Model A) to effective high-capacity learning (Models B and C). Importantly, improvements in accuracy are not accompanied by the characteristic signatures of overfitting, supporting the conclusion that architectural expressiveness, rather than training instability, explains the observed performance differences.

5 Generalization Test

5.1 External In-Distribution Evaluation

To disentangle generalization within the same acquisition domain from robustness to genuine domain shift, the best model (Model C) was additionally evaluated on the *rps-cv-images* dataset. This dataset is external to the training and validation splits, but shares similar characteristics in terms of background, lighting conditions, and gesture presentation.

Model C achieves near-saturated performance on this benchmark, with an overall accuracy of 0.9982 across 2,188 test images (4 misclassifications in total). The confusion matrix is almost perfectly diagonal, and all per-class precision, recall, and F1-scores exceed 0.997.

Detailed quantitative results, including the full classification report and confusion matrix, are provided in Appendix C.

These results indicate that the selected architecture generalizes extremely well to unseen data drawn from the same underlying distribution, with residual errors attributable to isolated sample-level ambiguities rather than systematic decision failures.

5.2 External Out-of-Distribution Evaluation

Custom Dataset description

The custom dataset used for the out-of-distribution evaluation consists of a small collection of hand gesture images representing the three target classes (23 images for *rock* and *scissors*, 22 images for *paper*), acquired outside the original training distribution. The images were captured using consumer-grade devices under unconstrained conditions and include hands belonging to multiple individuals rather than a single subject. This choice introduces natural variability in hand shape, skin tone, finger proportions, and gesture execution style, thereby increasing the heterogeneity of the dataset.

In addition to subject variability, the images exhibit substantial diversity in background, lighting conditions, and viewpoint. Backgrounds range from uniform surfaces to textured environments such as wooden floors, tiled areas, and indoor furnishings, while lighting conditions vary from diffuse natural light to indoor artificial illumination. The hand poses are not strictly canonical: gestures may be partially rotated, cropped, or captured at non-standard distances and angles. Accessories such as watches or wristbands are also present in some images, further increasing visual complexity.

Although the dataset is approximately class-balanced, it is intentionally small and uncurated, reflecting realistic acquisition conditions rather than controlled laboratory settings. As such, it is not intended to provide statistically robust performance estimates, but rather to serve as a qualitative stress test for model robustness and generalization under domain shift. The dataset allows for a focused analysis of how well models trained on clean, curated data transfer to real-world scenarios characterized by higher intra-class variability and reduced visual consistency.

Out-of-Distribution Evaluation on Custom Data

The evaluation on the custom hand-gesture dataset reveals a marked degradation in performance under out-of-distribution (OOD) conditions for all architectures. Despite the dataset being nearly class-balanced, all three models exhibit a strong predictive bias towards the *scissors* class, as evidenced by highly skewed mean predicted priors before recalibration (Appendix D, Table 8).

In particular, Model B collapses most severely, assigning nearly all probability mass to *scissors* (predicted prior $\approx$ [0.026, 0.004, 0.970]), while Model A shows a similar but slightly milder skew ($\approx$ [0.077, 0.079, 0.844]). Model C, although more stable in its final decisions, still exhibits a strong prior concentration on *scissors* ($\approx$ [0.030, 0.000, 0.969]), indicating that domain shift induces a systematic confidence imbalance across architectures.

This behaviour is reflected in the confusion matrices and per-class metrics. For Models A and B, *scissors* attains comparatively higher F1-scores than *rock* and especially *paper*, with *paper* being severely penalized (Table 4). Model B completely fails to recover the *paper* class (F1 = 0.000), highlighting the extreme fragility of the shallow architecture under distributional shift.

Table 4: *Per-class F1-score on the custom hand-gesture dataset (post-recalibration).*

Model	F1 (rock)	F1 (paper)	F1 (scissors)
A	0.286	0.071	0.455
B	0.176	0.000	0.421
C	0.449	0.333	0.421

Model C, while not immune to domain shift, substantially mitigates the severity of the collapse once probability recalibration is applied. It avoids degenerating into an almost single-class predictor and achieves the highest overall accuracy (0.412) and macro F1-score (0.401) among the three models (Table 5). Unlike Model B, it retains non-trivial discriminative ability across all classes, particularly improving recognition of *rock*.

Overall, these results indicate that increased representational capacity and architectural regularization improve relative robustness under real-world variability, but do not eliminate sensitivity to substantial

Table 5: *Custom hand-gesture dataset evaluation by model. Metrics are macro-averaged over classes (post-recalibration).*

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
A	0.324	0.277	0.319	0.271
B	0.279	0.192	0.275	0.199
C	0.412	0.467	0.409	0.401

domain shift. Even the most expressive architecture (Model C) exhibits confidence collapse at the probabilistic level, suggesting that OOD generalization remains fundamentally challenging for small-scale CNN models trained on visually homogeneous datasets.

6 Conclusion and Future Work

This project investigated how increasing architectural complexity in convolutional neural networks affects both predictive performance and robustness to distributional shift in image classification tasks, using rock–paper–scissors hand gesture recognition as a controlled case study. Through a systematic comparison of three CNN architectures trained under identical optimization settings, the analysis demonstrated a clear relationship between model capacity and predictive performance. While the baseline architecture exhibited signs of underfitting and limited discriminative power, progressively deeper and better-regularized models achieved near-saturated in-distribution performance, confirming the benefits of increased representational expressiveness for this task.

Beyond in-distribution evaluation, the study placed particular emphasis on robustness under domain shift by introducing an external test set composed of unconstrained, real-world hand gesture images. This generalization test revealed a substantial degradation in performance for all architectures. In particular, simpler models exhibited a pronounced collapse toward visually dominant classes, producing highly skewed predictive distributions and severely penalizing minority classes. Even the most expressive architecture showed a strong probabilistic bias under shift, although it retained comparatively better class discrimination at the decision level and achieved the highest macro-averaged performance among the three models.

These findings demonstrate that strong in-distribution validation performance does not guarantee robustness in real-world scenarios. High-capacity architectures mitigate—but do not eliminate—the effects of distributional shift, and probabilistic confidence may collapse even when top-1 accuracy remains comparatively stable. Explicit evaluation under out-of-distribution conditions is therefore essential for a realistic assessment of model reliability.

Despite these insights, several limitations must be acknowledged. The custom dataset used for the generalization test is intentionally small, preventing statistically robust performance estimates and limiting the strength of conclusions about real-world deployment. Moreover, all models were trained exclusively on a curated dataset, without advanced data augmentation, domain generalization techniques, or explicit calibration strategies that could further enhance robustness.

Future work could explore stronger augmentation pipelines, the inclusion of heterogeneous real-world data during training, and the use of pretrained backbone architectures to improve OOD generalization. Additionally, investigating probabilistic calibration methods and confidence-aware decision mechanisms may help address residual uncertainty under distributional shift.

Finally, from an application perspective, the final model could be deployed as a lightweight web application enabling interactive rock–paper–scissors play, while simultaneously serving as a demonstrative case study on the importance of robustness evaluation in practical computer vision systems.

Appendix

A Validation Confidence Diagnostics

Table 6: *Prediction confidence diagnostics on the validation set: mean and standard deviation of the maximum softmax probability across samples.*

Model	Val max-prob ($\mu \pm \sigma$)
Model A (best)	0.44 ± 0.09
Model B (best)	0.90 ± 0.14
Model C (best)	0.95 ± 0.06

B Additional Qualitative Diagnostics

B.1 Misclassified examples

Images are ranked by decreasing prediction confidence. Each panel reports the true label (T), the predicted label (P), and the associated softmax confidence score.

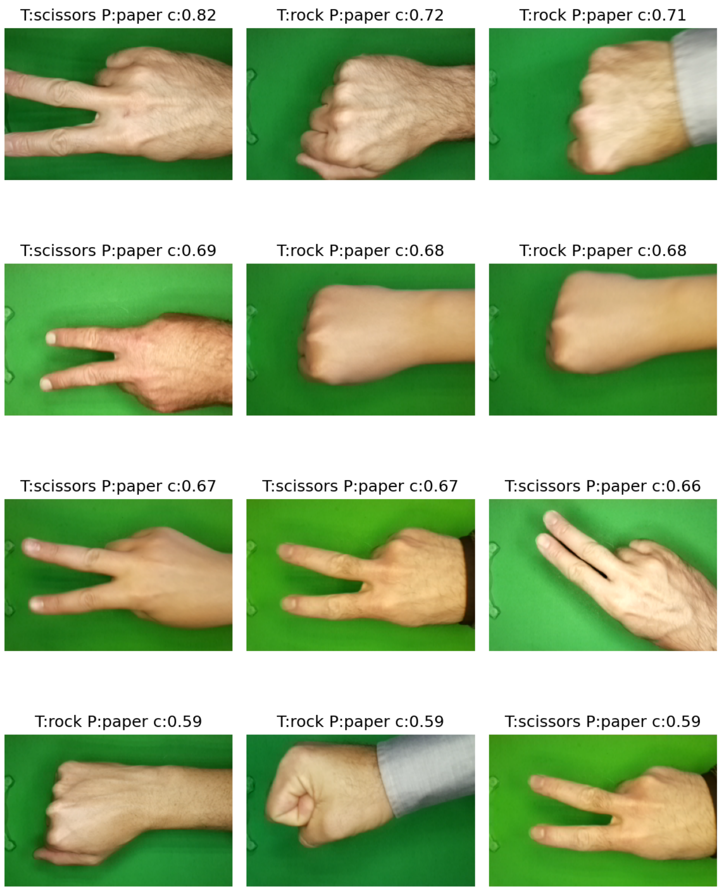

Figure 3: *Most confident misclassified validation samples for **Model A**.*

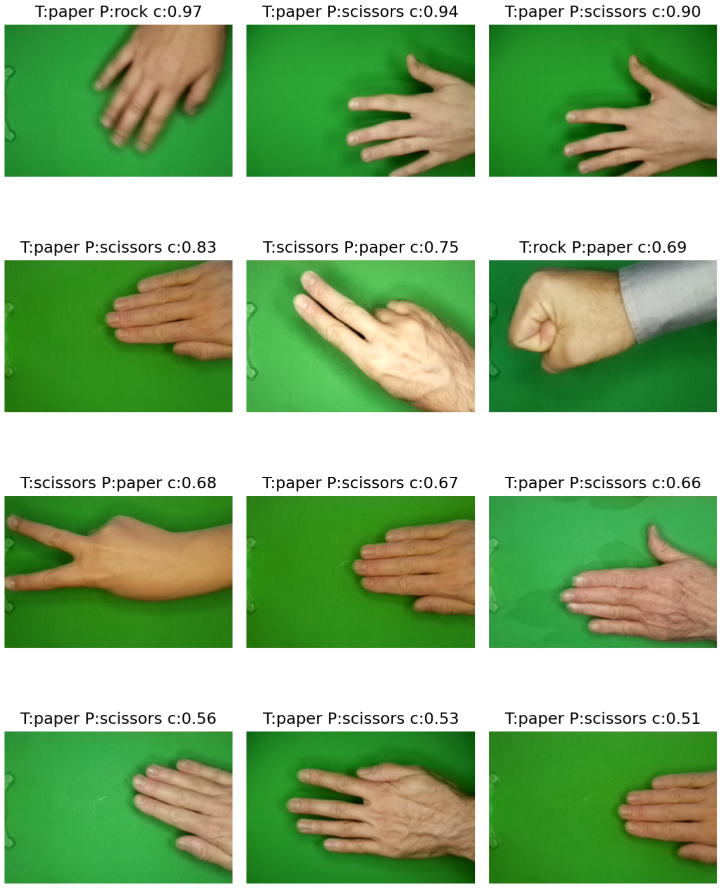

Figure 4: *Most confident misclassified validation samples for* ***Model B****.*

Figure 5: *Most confident misclassified validation samples for* ***Model C****.*

C External In-Distribution Evaluation Details

C.1 Classification report

Table 7: *Classification report for Model C on the external in-distribution test set* rps-cv-images.

Class	Precision	Recall	F1-score	Support
Rock	0.9972	0.9986	0.9979	726
Paper	0.9972	0.9972	0.9972	712
Scissors	1.0000	0.9987	0.9993	750
Accuracy			0.9982	2188
Macro avg	0.9981	0.9982	0.9982	2188
Weighted avg	0.9982	0.9982	0.9982	2188

C.2 Confusion matrix

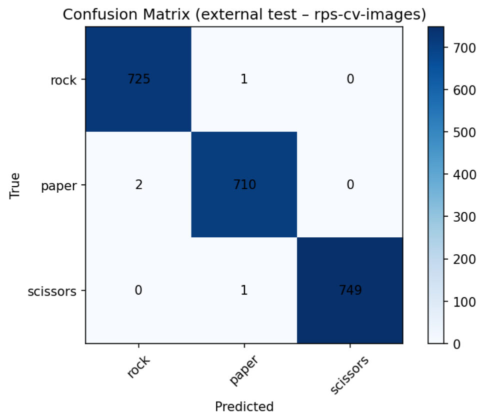

Figure 6: *Confusion matrix for model C on the external in-distribution test set (*rps-cv-images*).*

D External Out-Of-Distribution Evaluation Details

D.1 Confusion matrices

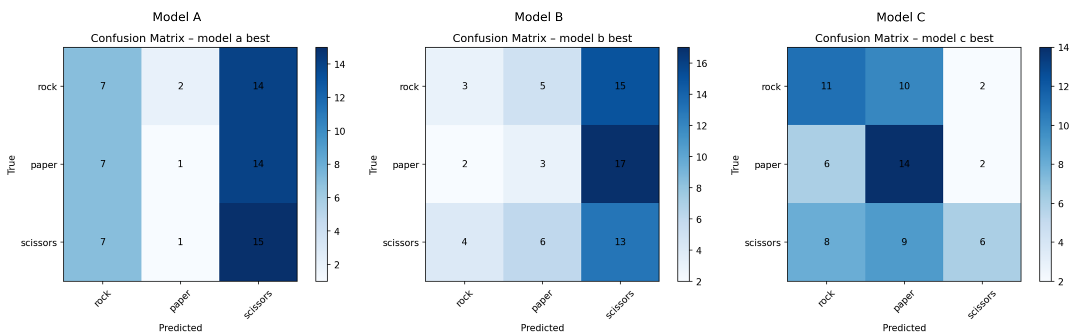

Figure 7: *Confusion matrices for the three architectures evaluated on the custom out-of-distribution dataset.*

D.2 Predicted class priors

Table 8: *Mean predicted class priors (pre-recalibration) for the three architectures on the custom out-of-distribution dataset.*

Model	**Rock**	**Paper**	**Scissors**
Model A	0.077	0.079	0.844
Model B	0.026	0.004	0.970
Model C	0.030	0.000	0.969
Target prior	0.333	0.333	0.333